Reg No.:_______________ Name:__________________________

# APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree Regular Examination June 2023 (2019 Scheme)

**Course Code: EET426**

**Course Name: SPECIAL ELECTRIC MACHINES**

**Max. Marks: 100** **Duration: 3 Hours**

## PART A

***Answer all questions, each carries 3 marks.*** Marks

| | | |
|---|---|---|
| 1 | Explain the significance of hall effect sensors in Brushless DC motors. | (3) |
| 2 | List any three applications of Permanent magnet Synchronous motor. | (3) |
| 3 | Define detent torque and holding torque of a stepper motor. | (3) |
| 4 | What are the factors on which the step angle of a single stack variable reluctance stepper motor depends? | (3) |
| 5 | Draw and explain split link circuit configuration power converter circuit for switched reluctance motor. | (3) |
| 6 | What are the advantages of Synchronous Reluctance motors? | (3) |
| 7 | Explain series split field DC servomotor with neat figure. | (3) |
| 8 | Why damping is necessary for AC servomotors? | (3) |
| 9 | Why compensating winding is used in AC series motor? | (3) |
| 10 | Compare Slotless Linear synchronous Motors and Slotted Linear Synchronous Motors. | (3) |

## PART B

***Answer any one full question from each module, each carries 14 marks.***

### Module I

| | | | |
|---|---|---|---|
| 11 | a) | Explain the principle of operation of Permanent magnet Brushless DC motors for $120^0$ commutation with necessary diagrams and truth tables. | (14) |

**OR**

12 a) Draw the block diagram for sensorless control of Permanent magnet Synchronous motor. (4)

b) Explain the construction and principle of operation of Permanent magnet DC motors with neat diagrams. (10)

**Module II**

13 a) Draw and explain the static and dynamic characteristics of stepper motor. (10)

b) Explain monofiliar and bifiliar windings. (4)

**OR**

14 a) Describe the various modes of operation of Hybrid stepper motor. (14)

**Module III**

15 a) Explain the principle of operation of 8/6 switched reluctance motor with neat diagrams. (14)

**OR**

16 a) Why rotor position sensor is required for the operation of switched reluctance motor? (4)

b) With neat diagram explain the construction of Synchronous Reluctance Motor? (10)

**Module IV**

17 a) Draw the Torque speed curves of an AC servomotor for various value of control voltage. (4)

b) Explain various types of rotor constructions for AC servomotors. (10)

**OR**

18 a) Derive the transfer function of the armature controlled DC Servomotor. (9)

b) Compare field controlled and armature controlled DC servomotor. (5)

## Module V

| | | | |
|---|---|---|---|
| 19 | a) | Develop the equivalent circuit of Linear Induction motor and describe main factors affecting its performance. | (10) |
| | b) | What are the advantages of linear motors? | (4) |

**OR**

| | | | |
|---|---|---|---|
| 20 | a) | Explain the constructional differences between Permanent Magnet Linear Synchronous motors with Active reaction rail and Permanent Magnet Linear Synchronous motors with Passive Reaction Rail. | (10) |
| | b) | Draw and explain torque -speed characteristics of Hysteresis Motor. | (4) |

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| **Sahrdaya College of Engineering and Technology, Kodakara** | | |
| **Answer Key** | | |
| **APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**<br>EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023 | | |
| **Course Code: EET 426** | | |
| **Course Name: SPECIAL ELECTRICAL MACHINES** | | |
| **PART A** | | |
| | | ***Answer all questions. Each question carries 3 marks*** |
| | ***Q1*** | ***Explain the significance of hall effect sensors in Brushless DC motors.*** |
| | Ans | In a BLDC motor, Hall effect sensors are employed to determine the position of the rotor accurately.These sensors are typically placed in the stator and are evenly spaced around it. They detect the magnetic field of the rotor's permanent magnets as they pass by, signaling the controller about the rotor's position.The information from Hall effect sensors is used by the motor controller to determine the right amount of current to the motor windings at the right time, optimizing performance and reducing energy consumption. This ensures that the magnetic field produced by the stator aligns with the permanent magnets on the rotor, creating a continuous rotation.Accurate rotor position feedback allows for precise control of the motor. This, in turn, enables smooth and efficient operation with minimal power losses. |
| | ***Q2*** | ***List any three applications of Permanent magnet Synchronous motor.*** |
| | Ans | a) Compressor and blowers<br>b) Steel rolling mills<br>c) Ship propulsion |
| | ***Q3*** | ***Define detent torque and holding torque of a stepper motor*** |
| | Ans | *Holding Torque*<br>It is the maximum load torque which can be applied to the shaft of the energized stepper motor without causing continuous rotation. If the holding torque is exceeded, the motor suddenly slips from the present equilibrium position and it will move one full step when the windings are energized.<br>*Detent torque*<br>It is the maximum load torque which the un-energized stepper motor can withstand without causing rotation. Detent torque is due to magnetism, and is therefore available only in permanent magnet and hybrid stepper motor. |

**Q4** **What are the factors on which the step angle of a single stack variable reluctance stepper motor depends?**

Ans

$$\text{Step angle, } \beta = \left(\frac{N_s - N_r}{N_s * N_r}\right) * 360^\circ$$

The step angle of a single stack variable reluctance stepper motor depends on several key factors:

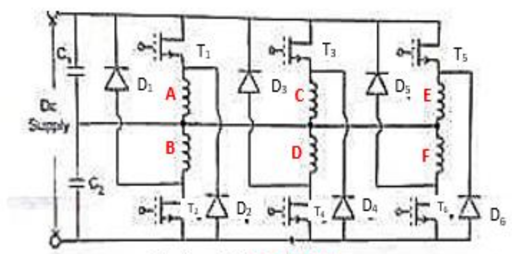
*Number of Stator Poles*: The step angle is influenced by the number of stator poles. More poles typically result in a smaller step angle. For example, a motor with 8 stator poles will have a smaller step angle compared to a motor with 4 stator poles.

*Number of Rotor Teeth:* The number of teeth on the rotor also affects the step angle. A rotor with more teeth will have a smaller step angle. Conversely, a rotor with fewer teeth will have a larger step angle.

*Voltage and Current Applied*: The voltage and current applied to the motor coils influence the magnetic field strength and, consequently, the motor's ability to move from one step to the next. Higher currents and voltages can lead to more precise and smaller step angles.

***Q5*** ***Draw and explain split link circuit configuration power converter circuit for switched reluctance motor.***

Ans




Six phase SRM (A,B,C,D,E,F)

The main power supply is split into two halves using split capacitors.

During conduction, energy is supplied to the phases by one half the power supply. During commutation period, the stored energy is fed back to other half of the power supply.When switch T1 is turned on, phase winding A is energized by capacitor C1. When switch T1 is turned off, the stored energy in the phase winding A is fed back to the capacitor C2 through diode D2.When T4 is turned on by capacitor C2 and phase winding D is energized. When switch T4 is turned off, stored energy in the winding D is feedback to the capacitor C1 through diode D3. The similar operation takes place in the remaining winding also.

***Q6*** ***What are the advantages of Synchronous Reluctance motors?***

Ans

*High Efficiency*

SynRM motors can achieve high efficiency levels, especially under partial load conditions. Their design allows for reduced losses, contributing to energy savings and improved overall efficiency.

*Simple and Robust Construction:*

Synchronous Reluctance Motors have a simpler and more robust construction compared to some other motor types, such as permanent magnet motors. This simplicity often leads to lower manufacturing costs and increased reliability.

*No Permanent Magnets:*

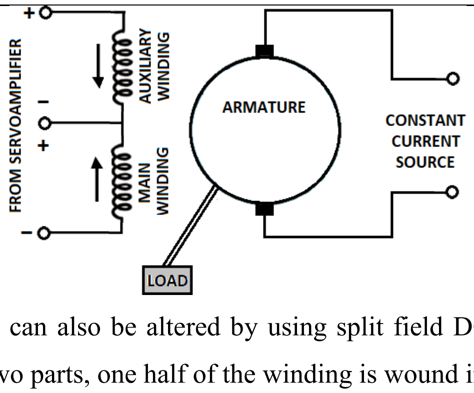
Unlike Permanent Magnet Synchronous Motors (PMSM), SynRM motors do not use permanent magnets. This can be an advantage in terms of cost, as permanent magnets can be expensive. Additionally, it eliminates concerns related to the supply and cost fluctuations of rare earth magnets.

*Suitable for High-Speed Applications:*

Synchronous Reluctance Motors can be well-suited for high-speed applications. Their construction allows for efficient operation at higher speeds, making them suitable for applications where high rotational speeds are required.

***Q7*** ***Explain series split field DC servomotor with neat figure.***

Ans



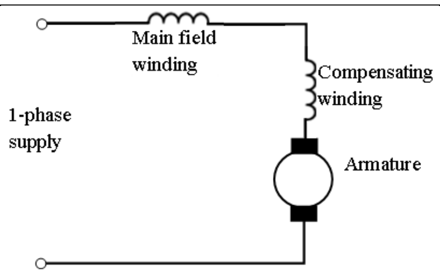

The direction of rotation can also be altered by using split field DC motor. where the field winding is divided into two parts, one half of the winding is wound in clockwise direction and other half in wound in anticlockwise direction The amplified error signal is fed to the junction point of these two halves of the field. The magnetic field of both halves of the field winding opposes each other. During operation of the motor, magnetic field strength of one half dominates other depending upon the value of amplified error signal fed between these halves. Based on that,the motor rotates in either clockwise or anticlockwise direction.

***Q8*** ***Why damping is necessary for AC servomotors?***

Ans Servomotors are electrical actuators used in feedback control system, which directly or indirectly,actuate something that influences the output being regulated by the system. The servomotor must respond rapidly to the change in control voltage. The motor must be stable, it must not oscillate or overshoot. Rapid response is obtained by having high torque to inertia ratio in order to have an initial rapid acceleration. Overshoot is minimized and stability is achieved using a damping or retarding torque that increases with rotational speed

***Q9*** ***Why compensating winding is used in AC series motor?***

Ans




In order to reduce the effect of armature reaction, thereby improving commutation and reducing armature reactance, a compensating winding is used. The compensating winding is put in the stator slots. The axis of the compensating winding is 90º (electrical) with the main field axis.

***Q10*** ***Compare Slotless Linear synchronous Motors and Slotted Linear Synchronous Motors.***

Ans

| Slotless LSM | Slotted LSM |
|---|---|
| Higher efficiency in higher speed range | Higher efficiency in lower speed range |
| Lower winding cost | Higher thrust density |
| Lower thrust propulsion | Lower input current |
| Lower acoustic noise | Less number of PMs |

**PART B**

***Answer any one full question from each module. Each question carries 14 marks***

**Module 1**

***Q11*** ***Explain the principle of operation of Permanent magnet Brushless DC motors for 120° commutation with necessary diagrams and truth tables.***

Ans Consider a three phase BLDC motor with two rotor poles and having an arc length of 180˚. The number of slot per pole per phase is two. Two adjacent coils in the armature forms a phase.

The coils are assumed to be full pitched and has N turns each. The slot pitch is 30˚ and single layer winding is used. The switching interval is 60˚.

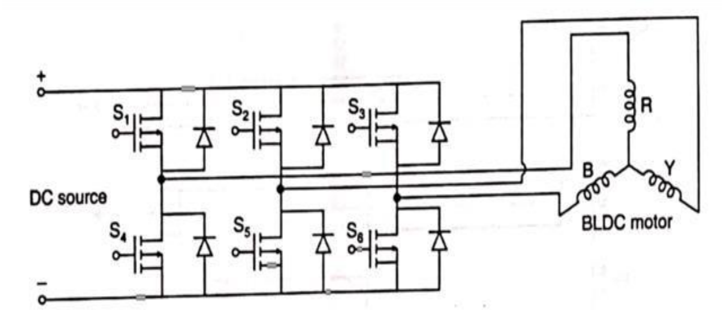
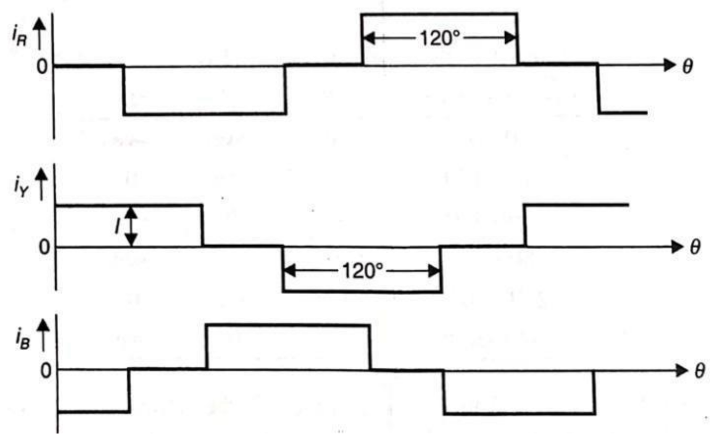


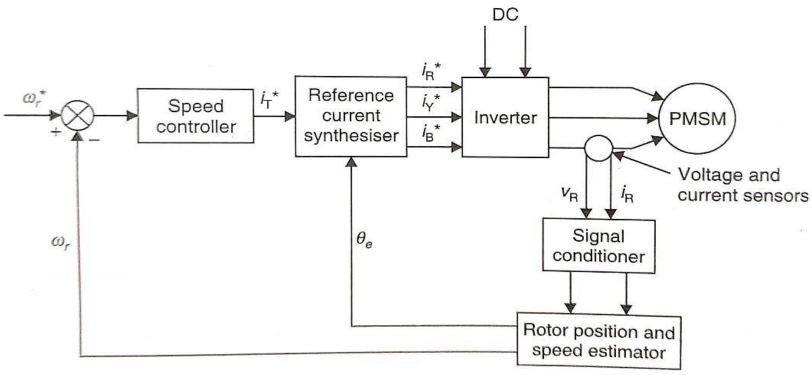

**Table** Switching interval and phase currents

| *Switching interval* | *R* | *Y* | *B* |
|---|---|---|---|
| 0°–60° | 0 | +ve | –ve |
| 60°–120° | –ve | +ve | 0 |
| 120°–180° | –ve | 0 | +ve |
| 180°–240° | 0 | –ve | +ve |
| 240°–300° | +ve | –ve | 0 |
| 300°–360° | +ve | 0 | –ve |

When ωt = 0 the phase Y is conducting positive current and phase B is conducting negative current. For that, the switches S2 and S6 are turned ON. The polarity of mmf distribution is same as that of the flux density distribution of rotor. So positive torque is produced. At ωt = 60˚ the switch S6 is switched OFF and S4 is turned ON. Now the torque remains unaffected. So the rotor continuous to rotate. For achieving the phase current pattern in the Table, the semiconductor switches are turned ON in the sequence S2, S6 ; S2, S4; S3, S4 ; S3, S5 ; S1, S5 ; S1, S6 . . . , while all other switches are in OFF position. The current waveforms are shown in Figure

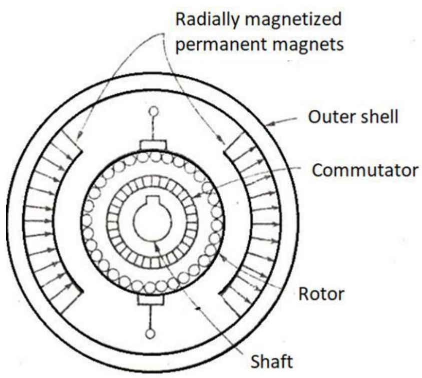


| | | |
|---|---|---|
| | ***Q12*** | ***a) Draw the block diagram for sensorless control of Permanent magnet Synchronous motor. (4)***<br>***b) Explain the construction and principle of operation of Permanent magnet DC motors with neat diagrams. (10)*** |
| | Ans | a) |




Block diagram of sensorless control scheme for PMSM

***b) Construction and principle of operation of Permanent magnet DC motors***




The field poles of this motor are essentially made of permanent magnet. A PMDC motor mainly consists of two parts.

-  Stator
-  Rotor

*Stator*

Stator is a steel cylinder and the magnets are mounted in the inner periphery of this cylinder. The steel cylindrical stator also serves as low reluctance return path for the magnetic flux. The permanent magnets are mounted in such a way that the N-pole and S-pole of each magnet are alternatively faced towards armature. Although field coil is not required in permanent magnet dc motor, sometimes they are used along with permanent magnet. This is because if permanent magnets lose their strength, these lost magnetic strengths can be compensated by field excitation through these field coils.

*Rotor*

The rotor of PMDC motor is similar to the armature other DC Motors. It is made of number of silicon steel sheets to reduce eddy current loss. Consists of core, windings, commutator and brushes. There are three types of armature structures available:

1. Slotted Armature
2. Slotless Armature
3. Moving Coil

*Principle of Operation*

The working of a PMDC motor is based on the principle that when a current-carrying conductor is placed in a magnetic field, it experiences a force whose direction is given by Fleming's left-hand rule and magnitude is given by $F = BIl$, where $B$ is the flux density, $I$ is the current through the conductor and $l$ is the length of the conductor. Consider a part of PMDC motor shown in Fig. 3.8.

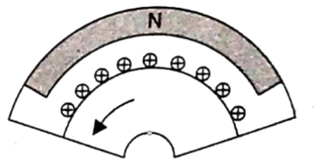



**Fig. 3.8** Part of a PMDC motor

Let the armature carry current in the direction shown in Fig. 3.8, when supply is switched ON. Due to the interaction between the current and the magnetic field, the conductor experiences a force. The force is in the direction shown. As the conductors are placed on the rotor, the rotor starts rotating in the CCW direction. Once the armature starts rotating, a back emf is produced in the armature winding. The direction of back emf is opposite to the applied voltage. The applied voltage has to force current through the armature conductors against this back emf. The electrical energy for overcoming this opposition is converted into mechanical energy developed in the armature.

**Module 2**

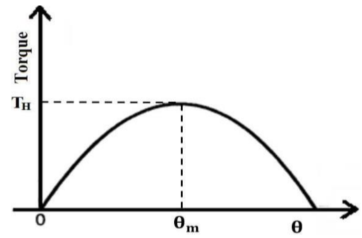
| | |
|---|---|
| ***Q13*** | ***a) Draw and explain the static and dynamic characteristics of stepper motor. (10)***<br>***b) Explain monofiliar and bifiliar windings. (4)*** |

Ans

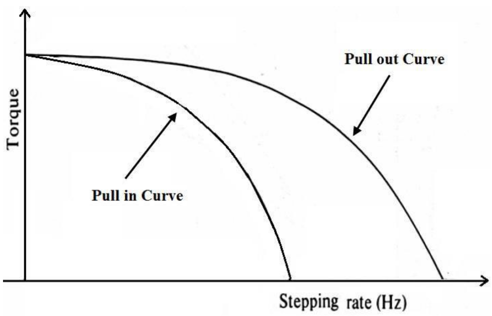
***a) Static Characteristics of Stepper Motor***

The stepper motor is excited and brought into rest or equilibrium position by supplying a current. If an external torque is applied to the shaft, an angular displacement will occur. The relation between the external torque and the displacement is shown in Figure. This curve is called Torque-Angle characteristics and the maximum of static torque is termed the holding torque which occurs at $\theta = \theta m$.

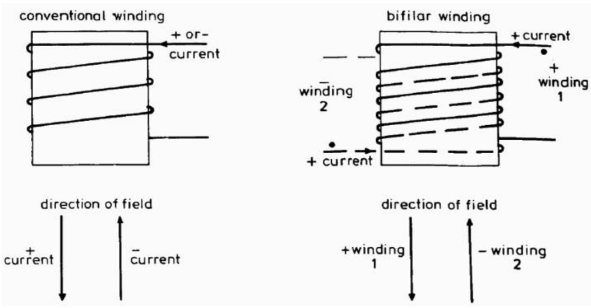
The holding torque is defined as the maximum static torque that can be applied to the shaft of an excited motor without causing continuous motion.



*Dynamic Characteristics of Stepper Motor*
The characteristics relating to motors which are in motion or about to start are called dynamic characteristics.
Pull-in torque and Pull-out characteristics of a stepper motor is called dynamic characteristics or torque-speed characteristics. The area between pull-in torque and pull-out torque represents those
values of torque and stepping rate which the motor follows without loosing synchronism provided that
it has already started and synchronised. This area is known as *slew range*.




***b) Monofiliar and bifiliar windings.***

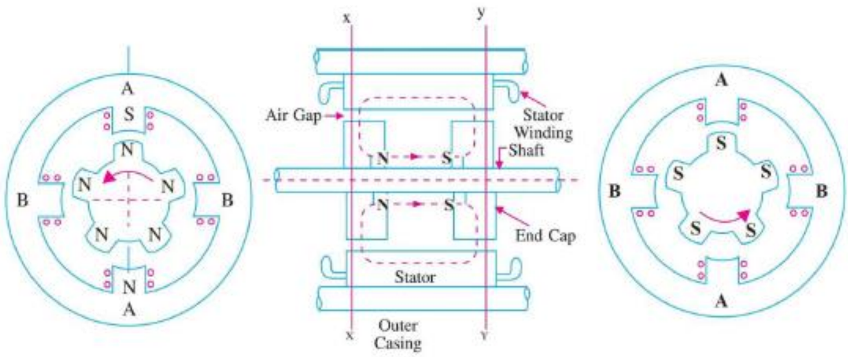



A bidirectional current flowing in the hybrid motor windings produces a bidirectional field in the stator poles. With a bifilar winding the same results is achieved by two pole windings in opposite senses,as illustrated for one pole in Fig.. Depending on the field direction, one of the windings is excited by a unidirectional current; in Figurethe field produced by a positive current in the conventional arrangement is available by exciting the bifilar +winding with positive current. The effect of negative current in the conventional winding is then achieved by positive excitation of the bifilar winding.
Simple ON/OFF switching is adequate for bifilar winding while current reversal circuitry is required for unifilar winding

+ Since current reversal is absent in bifilar winding, there are smaller levels of induced voltages by self induction and mutual induction. For this reason, the dynamic torque at a given stepping rate is usually larger for bifilar stepper motors, particularly at higher speed.

+ Bifilar motors produce large torque compared to monofilar motors.

+ Bifilar wound hybrid motors, can be operated more efficiently with a unipolar drive.

| | Q14 | ***Describe the various modes of operation of Hybrid stepper motor.*** |
|---|---|---|
| | Ans | Hybrid stepper motors combine features of both Permanent Magnet (PM) and Variable Reluctance (VR) stepper motors. They offer several modes of operation, providing flexibility and performance advantages for various applications. Here are the primary modes of operation for hybrid stepper motors:<br>*One-Phase On (Wave Drive):*<br>In this mode, only one of the two phases (usually the A-phase or U-phase) is energized at a time.The rotor aligns itself with the magnetic field of the energized phase, taking a single step.This mode provides relatively low torque and is typically used in applications where power efficiency is not a primary concern.<br><br><br>*Two-Phase On (Full Step):*<br>In this mode, both phases (A and B or U and V) are energized simultaneously with equal current.The rotor aligns itself between the two phases, resulting in a step angle that is half of the full step angle.Two-phase on mode provides higher torque than one-phase on mode but may require more current and produce more heat.<br>Half-Step Mode:<br>In the half-step mode, the phases are alternated between one-phase on and two-phase on states. During each step, the rotor moves half of the full step angle. Half-stepping provides finer resolution and smoother motion compared to full stepping but may have lower torque compared to full step mode.<br>*Microstepping:*<br>Microstepping is a more advanced mode of operation that allows for very fine control of the rotor position. It involves subdividing each full step into smaller steps. Microstepping requires a special microstepping driver/controller and can provide hundreds or even thousands of steps |

per full step. This mode offers significantly smoother motion, reduced vibration, and lower audible noise. It also reduces the risk of resonance effects. Microstepping comes at the cost of reduced torque per step compared to full or half-step modes.

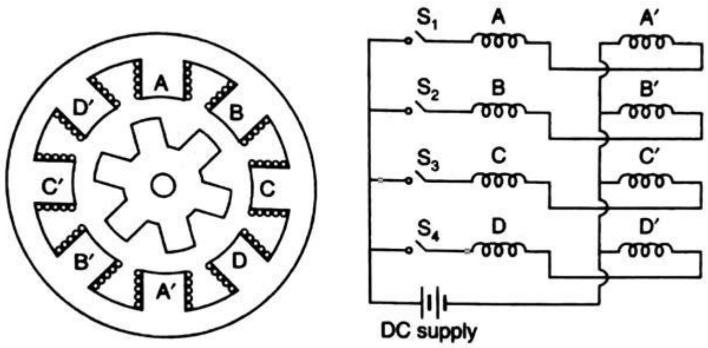
**Module 3**

***Q15*** ***Explain the principle of operation of 8/6 switched reluctance motor with neat diagrams.***

Ans ***Principle of Operation of 8/6 SRM***

The operation principle is based on the variation in magnetic reluctance. The rotor always try to align with low reluctance path. When a stator coil is excited, the rotor experiences a force which will pull the rotor to the aligned position. 8/6 SRM has 4 phases A-A', B-B', C-C' and D-D'. These phases are excited by DC supply through switches S1, S2, S3 and S4 as shown in Figure 3.7. When a stator coil is excited, the rotor experiences a force which will pull the rotor to the aligned position.

When phase A-A' is energized, the rotor comes to the maximum inductance (minimum reluctance) position as in the first case. The rotor teeth 1 and 4 are attracted to the stator teeth A and A' respectively as shown in Figure (a).They are in the minimum reluctance position as far as phase windings is concerned. Then (dLa / dq) = 0). At this position inductance of B windings is neither maximum nor minimum.

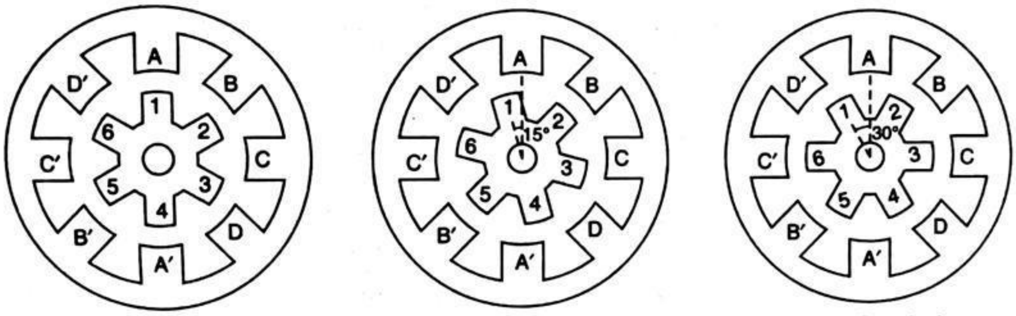



**Four-phase, eight-pole SRM**

Next, the switch S2, is turned ON and switch S1, is turned OFF. Now the phase A is de-energized and the phase B is energized so that the stator teeth 3 and 6 are energized. The stator teeth of B and B' would align with rotor teeth 2 and 5 respectively as shown in Figure (b). The rotor rotates in the counter clockwise (CCW) direction and makes an angular displacement of 15˚. The torque developed is equal

$$\frac{1}{2} i_B^2 \left(\frac{dL_A}{d\theta}\right)$$

If this torque is more than the opposing load torque and frictional torque the rotor starts rotating




Phase A excited (a) | Phase B excited (b) | Phase C excited (c)

Now phase winding B is switched OFF and phase winding C is turned ON to DC supply. Then the rotor experiences a torque as $\left(\frac{dL_B}{d\theta}\right)$ exists. The rotor continues to rotate. When the rotor rotates further 15°, the torque developed due to winding C is shown in Figure (c) Then the phase winding C is switched off and phase winding D is energized. Then rotor experiences a torque and rotates further step 15°. This is a continuous and cyclic process. Thus the rotor starts. It is a self-starting motor. If the switching sequence is changed to $S_1, S_4, S_3, S_2, S_1, S_4 \ldots$, then the direction of rotation is reversed.

***Q16*** ***a) Why rotor position sensor is required for the operation of switched reluctance motor? (4)***

***b) With neat diagram explain the construction of Synchronous Reluctance Motor? (10)***

Ans

a) A rotor position sensor is essential for Switched Reluctance Motors to enable precise control, optimize performance, and ensure reliable operation, especially in applications with varying loads and operating conditions.A rotor position sensor is required for the operation of a Switched Reluctance Motor (SRM) for several important reasons:

- Efficient Control and Commutation:
- Improved Efficiency and Performance:
- Start-Up and High-Speed Operation:
- Fault Detection and Diagnostics:
- Closed-Loop Control:

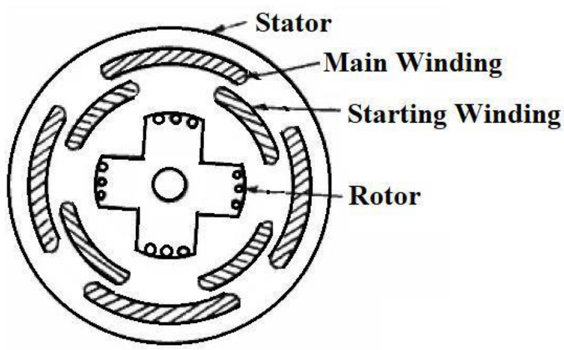
Closed-loop control enhances the motor's precision, accuracy, and ability to adapt to changing loads or operating conditions.

Common types of rotor position sensors used in SRMs include:

*Hall Effect Sensors*: These sensors can be mounted on the stator and detect the position of the rotor as it passes by. They provide discrete information about the rotor's location.

*Optical Encoders*: Optical encoders offer high-resolution position feedback by generating a continuous stream of pulses as the rotor rotates. They provide precise position information but may require more complex interfacing and electronics.

***b) Construction of Synchronous Reluctance Motor***

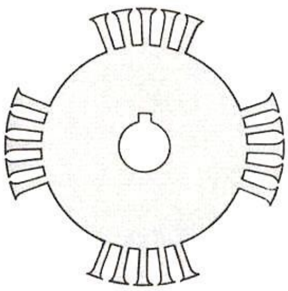


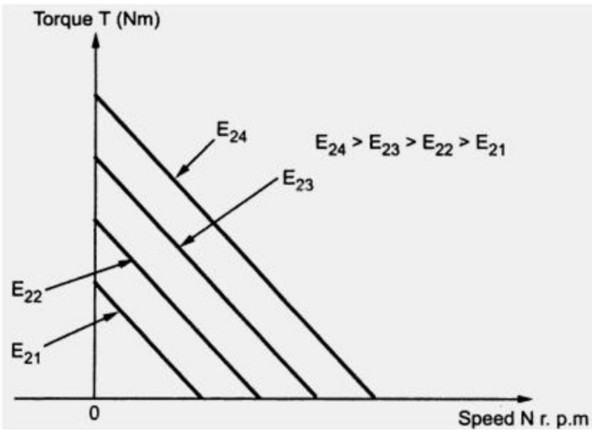

***Stator***

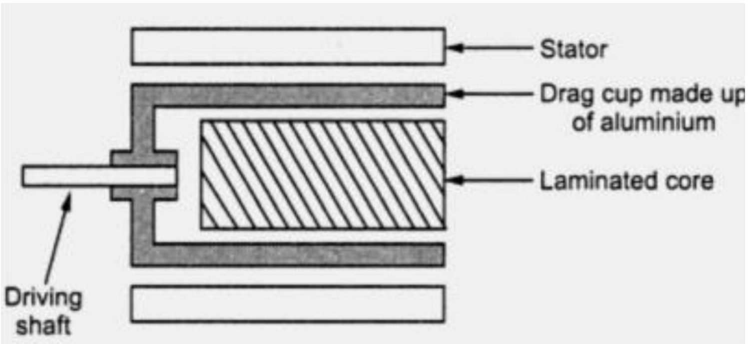
Stator is made up of laminations of ferromagnetic steel sheets. Slots for housing armature conductors are provided along the inner periphery of the armature. The stator has three phase distributed winding which creates rotating magnetic field in the air gap.

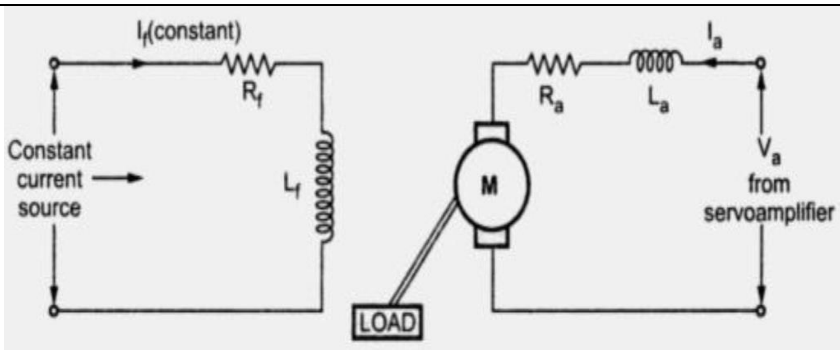
The stator of the single phase reluctance motor is similar to that of stator of the single phase induction motor. It consists of starting (auxiliary winding) and running winding (main winding) in the stator slots. This type of motor is also called as Split phase reluctance motor. There is a phase difference between the current and the corresponding fluxes in the starting and running winding. The interaction between these two fluxes produce rotating magnetic field.

**Rotor**



| | | |
|---|---|---|
| | | The rotor of the reluctance motor is of salient or projecting poles. No dc supply is provided to the rotor. The rotor of the reluctance motor is squirrel cage with some rotor teeth removed in certain places to provide desired number of salient rotor poles. In Figure rotor teeth are removed in four places to produce four pole rotor structure. So the air gap is not uniform. The reluctance depends on the air gap length. More the air gap, more is the reluctance. When the rotator rotates, due to the non-uniform air gap, reluctance also changes. In a synchronous reluctance motor, the stator and rotor should be constructed in such a way that the armature inductance varies sinusoidally |
| **Module 4** | | |
| | ***Q17*** | ***a) Draw the Torque speed curves of an AC servomotor for various value of control voltage. (4)***<br>***b) Explain various types of rotor constructions for AC servomotors. (10)*** |
| | Ans | **a)** ***Torque speed curves of an AC servomotor***<br><br> |
| | | ***b) Types of rotor constructions for AC servomotors***<br>The rotor is generally of two types. The two types of rotors are<br>+ Squirrel cage rotor<br>+ Drag cup type rotor<br>*Squirrel cage rotor*<br>The Squirrel cage rotor have large length and small diameter so its resistance is very high. Its air gap is kept small. The resistance of drag type rotor is very high and hence have high starting torque. The rotor consists of a cylindrical laminated core with parallel slots for carrying the rotor conductors. The rotor has small number of slots than the |

| | | |
|---|---|---|
| | | stator and must be a non-integral multiple of stator slots, so as to prevent magnetic locking of rotor. Each slot contains uninsulated heavy bars of copper, aluminium or alloys. The rotor conductors are short-circuited at the end by a end-rings of copper or aluminium or alloys. The rotor conductors and its end-rings forms a closed circuit which resembles a squirrel cage.<br>*Drag Cup rotor*<br><br><br>To reduce the inertia, a drag cup type of rotor construction is used. It is in the form of a hollow cylinder with one end open. As it is very thin and light, it has very small inertia, and considerable torque can be obtained. There are two air gaps in this construction. Compared to squirrel cage rotor, drag cup rotor has higher effective airgap. A stationary iron core is placed between the conducting cups to complete the magnetic circuit. The drag cup is made up of non-magnetic material like copper, aluminium or an alloy. The slotted rotor laminations in this construction. These are wound for as many number of poles as possible so that operating speed of motor is very low. Such a construction is used in very low power applications. The resistance of drag type rotor is very high and hence have high starting torque. |
| | ***Q18*** | ***a) Derive the transfer function of the armature controlled DC Servomotor.***<br>.<br>***b) Compare field controlled and armature controlled DC servomotor*** |
| | Ans | ***a) Transfer function of the armature controlled DC Servomotor*** |




In a armature controlled d.c. servomotor, the control signal available from the servoamplifier is applied
to the armature of the motor. The field winding is supplied with constant current hence the flux remains constant. So these motors are also called as constant magnetic flux motors. It is easy to keep field current constant rather than armature current. This is because presence of back e.m.f. in the armature

Here, $R_a$ = Resistance of armature in ohms

$L_a$ = Inductance of armature windings in H

$I_a$ = Armature current in A

$I_f$ = Field current in A

e = Applied armature voltage in V

$e_b$ = back emf in V

$T_a$ = Torque developed by the motor in Nm

J = Equivalent moment of inertia of motor and load referred to the motor shaft in $kgm^2$

B = Equivalent viscous friction coefficient of inertia of motor and load referred to the motor shaft in Nm/ (rad/s)

For DC motors $\phi \propto I_f$

$\phi = k_f I_f$, where $k_f$ is a constant ---(1)

also, $T_m \propto \phi I_a$

$I_m = k_l \phi I_a = k_i k_f I_f I_a$ ---(2)

In armature controller DC motor, the field current is kept constant.

$\therefore T_m = k_t I_a$ ($k_t$ = constant) ---(3)

Motor back emf $e_b \propto \frac{d\theta}{dt}$

$e_b = k_b \frac{d\theta}{dt}$ ($k_b$ = back-emf constant) ---(4)

The differential equation for the armature circuit is

$e = I_a R_a + L \frac{dI_a}{dt} + e_b$

Torque equation $= T_m = \frac{Jd^2\theta}{dt^2} + \frac{Bd\theta}{dt}$ ---(6)

Equating the eq (3) and (6)

$\frac{Jd^2\theta}{dt^2} + \frac{Bd\theta}{dt} = k_t I_a$ ---(7)

Taking Laplace transforms of equation (4), (5), (6), (7) we get

$E_b(s) = k_b s\theta(s)$ ----(8)

$(SL_a + R_a)\, I_a(s) = E(s) - E_b(s)$ ---(9)

$(Js^2 + Bs)\, \theta(s) = T_m(s) = k_t\, I_a(s)$ ---(10)

From above equation

$G(s) = \frac{\theta(s)}{E(s)} = \frac{k_t}{s[CR_a + sL_a](J_s + B) + k_t k_b}$

**Comparison Between armature Controlled and Field Controlled D.C. Servomotors**

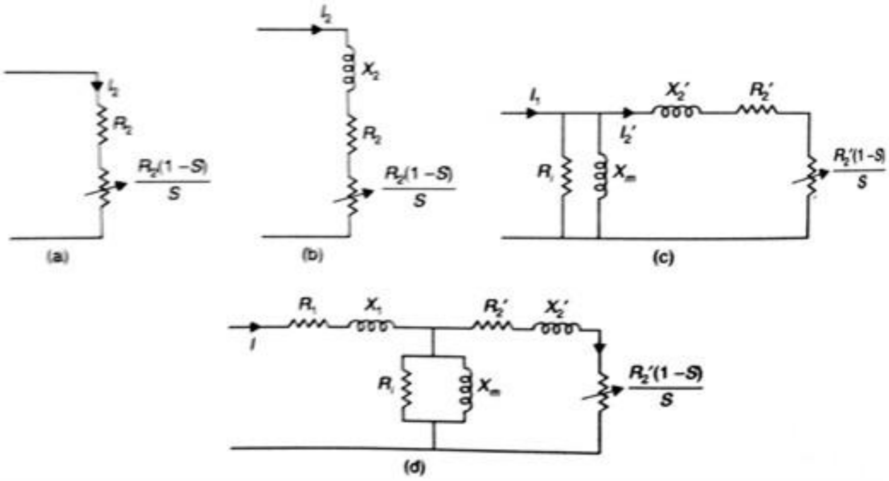
| Field Controlled | Armature Controlled |
|---|---|
| 1) Due to low power requirement amplifiers are simple to design. | 1) High power amplifiers are required to design |
| 2) Control voltage is applied to the field. | 2) Control voltage is applied to the armature. |
| 3) Time constant is large. | 3) Time constant is small. |
| 4) This is open loop system. | 4) This is closed loop system. |
| 5) Armature current is kept constant. | 5) Field current is kept constant. |
| 6) Poor efficiency. | 6) Better efficiency. |
| 7) Suitable for small rated motors. | 7) Suitable for large rated motors. |
| 8) Costly as field coils are must. | 8) Permanent magnet can be used instead of field coils which makes the motor less expensive. |

**Module 5**

**Q19** ***a) Develop the equivalent circuit of Linear Induction motor and describe main factors affecting its performance. (10)***

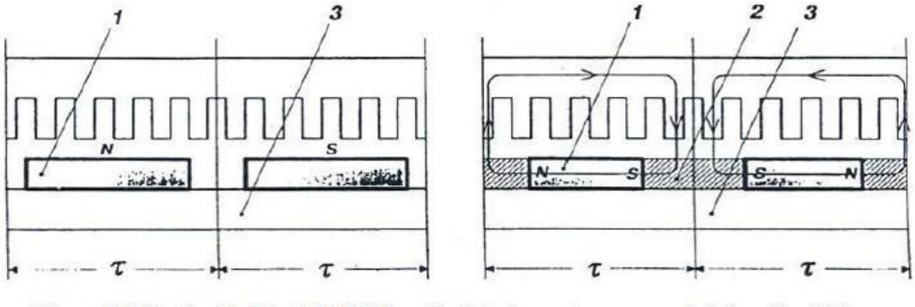
***b) What are the advantages of linear motors? (4)***

Ans a) ***The equivalent circuit of Linear Induction motor***

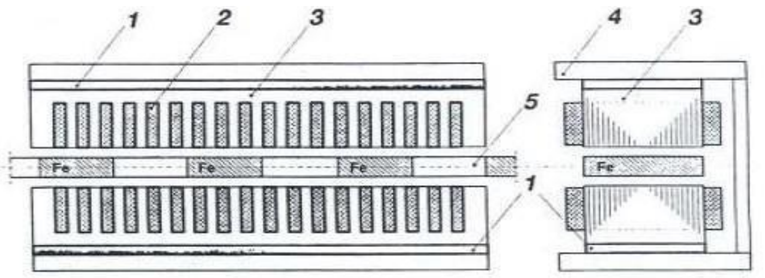


*Transverse edge effect and end effect*

Linear motors are associated with 2 effects which are absent in conventional induction motor. These effect will increase loss in motors

| | | |
|---|---|---|
| | | 1. Transverse edge effect<br>• Paths of induced current in secondary is not well defined because secondary of linear motor is a solid conducting plate<br>• The portion of current paths parallel to direction of motion of secondary does not make any contribution towards production of useful thrust but contribute towards losses<br>• This effect reduces effective thrust and increases losses and is known as transverse edge effect because current paths parallel to the direction of motion are more towards direction of conducting plates<br>2. End effect<br>In case of linear induction motor with short primary the current path towards end of field structure on the conducting plate go beyond the field structure and such portions of current path do not contribute to useful thrust but only towards motor losses.This is called end effect<br>***b) Advantages of Linear Motors***<br>● Low initial cost<br>● Low maintenance cost(absence of rotating parts)<br>● Simplicity<br>● No limitation on tractive effort due to adhesion between wheel and rail<br>● No limitation on maximum speed<br>● No overheating of rotor as the motor moves continuously over the cool stationary part leaving the heated portion behind.<br>● Better power to weight ratio |
| | ***Q20*** | ***a) Explain the constructional differences between Permanent Magnet Linear Synchronous motors with Active reaction rail and Permanent Magnet Linear Synchronous motors with Passive Reaction Rail. (10)***<br>***b) Draw and explain torque -speed characteristics of Hysteresis Motor. (4)*** |
| | Ans | ***a) Permanent Magnet LSM with Active Reaction Rail***<br>In surface arrangement of PMs the yoke (back iron) of the reaction rail is ferromagnetic and PMs are magnetized in the normal direction (perpendicular to the active surface). Buried PMs are magnetized in the direction of travelling magnetic field and the yoke is non- ferromagnetic, e.g. made of aluminum.<br><br><br>**Figure 5.8** Single sided flat PM LSMs with slotted armature core and: (a) surface PMs, (b) buried PMs. 1 -PM, 2 – mild steel pole, 3 - yoke. |

*PM with passive reaction rail*




Double sided homopolar PM LSM with passive reaction rail.1 - PM, 2 - armature winding, 3 – armature stack, 4 - yoke, 5.Reaction rail.

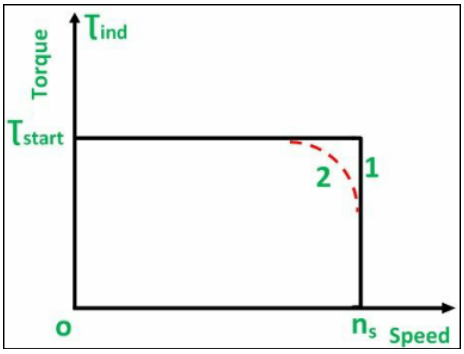
It has 2 polyphase armature windings. Both armature systems are magnetically and mechanically coupled by a ferromagnetic U-type yoke.

Each armature has a slotted stack with polyphase winding and PM are placed between stack and yoke. PM may be replaced by electromagnets.

The interaction between travelling magnetic field produced by armature winding and magnetic poles of reaction rail produces the thrust.

**b) *Torque Speed characteristic of Hysteresis Motor***

The speed torque curve of the motor is shown below.




Curve 1 is the ideal curve, and the curve 2 is the practical hysteresis motor curve. The torque-speed characteristic of the hysteresis motor is different from an induction motor. Since, at the synchronous speed, the torque developed by an induction motor becomes zero, whereas in the hysteresis motor the torque is constant at all the speed even at the synchronous speed. Thus, from the curve, it is seen that the locked rotor, starting and pull out torque is equal.

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